

Memory-Defined Phase Maps for Dissipative Quantum Reservoirs

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Abstract—We study when a stateful dissipative gate-model quantum reservoir is useful for time-series prediction. The main claim is not quantum advantage, but a reproducible phase-map claim: across tasks and seeds, good validation performance concentrates in a memory-defined operating regime with gentle input drive, nonmaximal recurrent coupling, and controlled amplitude damping. Memory capacity and intrinsic processing-capacity diagnostics identify this regime before exhaustive task sweeps. As supporting evidence, the same four-qubit, two-layer QRC96 architecture, with only ridge readout training, is competitive with an audited ESN100 baseline at matched classical readout dimension. Under shared validation selection it improves seed-level mean holdout NMSE, and under task-wise validation-plateau selection it improves the non-floor NARMA10 and Sunspots comparisons. The comparison is matched in readout features, not in measurement, shot, calibration, wall-clock, or hardware cost.

Index Terms—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, echo-state networks

I. QUESTION AND CLAIM

Reservoir computing trains a linear readout on fixed non-linear dynamics [1], [2]. In a quantum reservoir, the useful resource is not Hilbert-space dimension alone. The physical dynamics must preserve recent input history, forget old history, and expose nonlinear observables to a classical readout [4], [5], [6]. Excessive input drive overwrites the carried state; absent dissipation prevents controlled fading memory; absent recurrent mixing leaves weak temporal features.

This paper therefore treats the phase diagram as the scientific object. We ask whether a dissipative QRC has a transferable operating regime, rather than whether one tuned coordinate wins one benchmark. The main result is that phase maps generalize as a frequency-level regime across four tasks and ten reservoir seeds. Accuracy against ESN100 is a supporting claim: it tests whether features from this regime are useful at a comparable classical readout dimension.

II. MODEL AND SELECTION PROTOCOL

The reservoir carries a density matrix from one input time step to the next. For scalar input u_t ,

$$\rho_{t,0} = U_{\text{in}}(u_t)\rho_{t-1,L}U_{\text{in}}^\dagger(u_t), \quad U_{\text{in}}(u_t) = \bigotimes_i R_y(\beta u_t). \quad (1)$$

Each virtual layer applies frozen local R_x rotations, a ZZ ring, and product amplitude damping:

$$\rho_{t,\ell} = \mathcal{D}_\gamma^{\otimes n} \left[U_\lambda U_{\text{mix}} \rho_{t,\ell-1} U_{\text{mix}}^\dagger U_\lambda^\dagger \right], \quad U_\lambda = \prod_{(i,j) \in E_{\text{ring}}} e^{-i\lambda Z_i Z_j}. \quad (2)$$

The QRC96 architecture is fixed throughout the ESN comparison: $n = 4$, two layers, ring topology, R_x -ZZ- R_x mixer, uniform input, amplitude damping, and ridge readout only. QRC96 measures 12 local Pauli expectations and 36 ring-pair Pauli correlations per layer, giving 96 readout features.

For task j and seed s , let $r_{j,s}(\theta)$ be the validation percentile rank of $\theta = (\beta, \lambda, \gamma)$, with lower rank better. A transferable operating band is

$$B_{p,q} = \{\theta : \Pr[r_{j,s}(\theta) \leq p] \geq q\}. \quad (3)$$

This frequency-level definition avoids overclaiming a universal optimum. The strict all-seed top-20% intersection is empty, but $B_{20,0.7}$ contains four connected points and $B_{20,0.6}$ contains fourteen.

III. EVIDENCE

Phase-map transfer. Fig. 1 shows the core observation: low validation error is not scattered randomly across the grid. It concentrates in a small-input, nonzero-damping, nonmaximal-coupling region, and the leave-one-task-out maps preserve the same shape. The leave-one-task-out held-out validation rank is 0.240 with bootstrap CI [0.218, 0.271], so the regime is a top-quartile search prior rather than a task-specific coordinate.

Diagnostic mechanism. Memory capacity predicts the useful region. On the current QRC diagnostic rerun, MC has Spearman $\rho_s = -0.85$ against validation rank; IPC_{mem} , IPC_{tot} , and $\text{IPC}_{\text{nonlin}}$ give -0.84 , -0.90 , and -0.80 . Feature variation and effective rank are weakly positive, 0.16 and 0.15, so raw diversity alone retains poor reservoirs. Removing the mechanism also breaks accuracy: no recurrent mixing, zero damping, or dephasing raises holdout NMSE to near one, while QRC96 inside the regime is 0.077.

ESN comparison as supporting evidence. Under shared validation selection, QRC96 selects $(\beta/\pi, \lambda/\pi, \gamma) = (0.015, 0.100, 0.080)$ and ESN100 selects $(\rho, \text{input}, \text{leak}) = (0.9, 0.1, 1.0)$. Mean holdout NMSE is 0.077 for QRC96 and 0.090 for ESN100. The paired seed-level delta $\Delta = \text{NMSE}_{\text{ESN100}} - \text{NMSE}_{\text{QRC96}}$ is 0.0134 with CI [0.0082, 0.0187], and QRC96 wins 10/10 seeds. This aggregate

TABLE I
REVIEWER AUDIT TRAIL. HOLDOUT DATA ARE EXCLUDED FROM REGIME DISCOVERY AND HYPERPARAMETER SELECTION.

Item	Value
Tasks and splits	Mackey–Glass, Lorenz, NARMA10: washout/train/validation/holdout = 300/1600/400/400; annual Sunspots: 20/150/50/60. Inputs are scaled to $[-1, 1]$ and targets standardized.
Seeds and ridge	Reservoir seeds 42–51; ridge grid $\{10^{-8}, 10^{-7}, \dots, 10^3\}$ selected on validation.
QRC regime discovery	QRC16 coarse grid over $\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, $\gamma = \{0, 0.05, 0.12, 0.22, 0.30\}$; holdout unused.
QRC96 refinement	Same architecture; local grid $\beta/\pi = \{0.005, 0.010, 0.015, 0.020, 0.030\}$, $\lambda/\pi = \{0.035, 0.050, 0.075, 0.100, 0.125\}$, $\gamma = \{0.050, 0.080, 0.100, 0.120, 0.160\}$. The Sunspots audit adds a same-architecture fine grid inside the regime: $\beta/\pi = 0.006:0.001:0.016$, $\lambda/\pi = \{0.025, 0.030, 0.035, 0.040, 0.045, 0.050, 0.060, 0.075, 0.090, 0.100\}$, $\gamma = \{0.070, 0.080, 0.085, 0.090, 0.095, 0.100, 0.105, 0.110, 0.120\}$.
ESN100 audit	Dense Gaussian recurrent matrix scaled to spectral radius $\{0.5, 0.9, 1.1\}$, input scaling $\{0.1, 0.5, 1.0\}$, leak $\{0.3, 0.7, 1.0\}$, Gaussian bias std. 0.01, tanh activation, same tasks, splits, seeds, and ridge grid.
Selectors and tests	Shared selector: one setting minimizing mean validation NMSE across tasks and seeds. Task-wise selector: keep settings within 1% of best mean validation NMSE over seeds and choose the normalized hyperparameter medoid. Paired bootstrap CIs, Wilcoxon signed-rank tests, and sign tests use matched seed/task keys.
Resource scope	QRC96 and ESN100 are compared at matched classical readout dimension, 96 vs. 100 features. Quantum measurement bases, shots, calibration, wall-clock cost, and hardware noise are not modeled.

Phase maps reveal a transferable low-input, nonzero-damping operating regime ($\gamma = 0.12$)

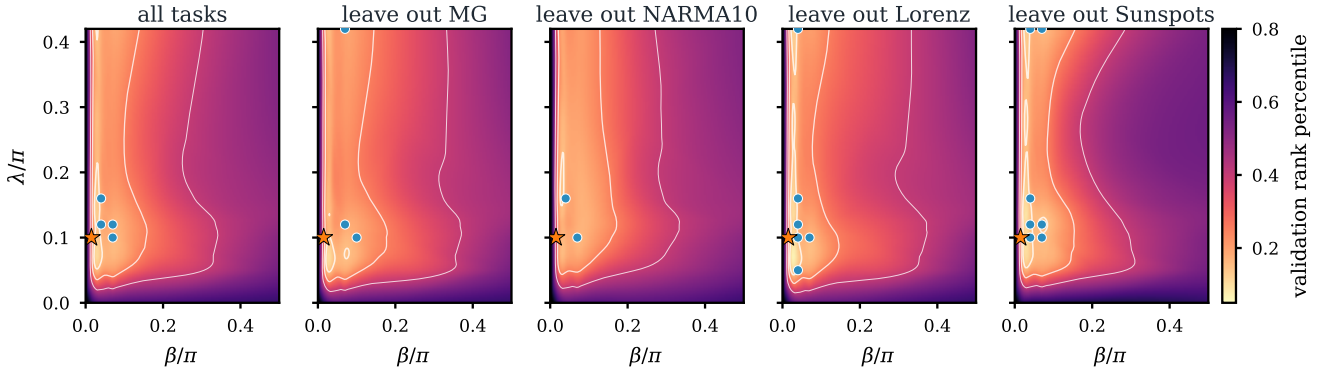


Fig. 1. Validation phase maps at the fixed $\gamma = 0.12$ slice. Bright regions are low validation percentile rank. Blue dots show coordinates that appear in the top-20% for at least 70% of task-seed replicates in the displayed conditioning set; the star is the shared QRC96 selected point's (β, λ) projection. The same low-input, nonmaximal-coupling regime persists under leave-one-task-out views.

is driven by NARMA10 under the shared selector, so we do not claim uniform task-wise dominance.

The task-wise validation-plateau selector keeps the architecture fixed but allows one setting per task for both QRC96 and ESN100. On the two non-floor tasks, QRC96 improves the combined task-seed mean by 0.0580 with CI $[0.0358, 0.0814]$ and wins 18/20 pairs. NARMA10 gives $\Delta = 0.0918$ with 10/10 wins. Sunspots gives $\Delta = 0.0242$, CI $[0.0026, 0.0565]$, 8/10 wins, and Wilcoxon $p = 0.0420$. Mackey–Glass and Lorenz are floor checks: ESN100 is already at numerical floor and is lower there.

ROBUSTNESS AUDIT: POSITIVE Δ FAVORS QRC96	
Shared seed means	$\Delta = 0.0134$, CI $[0.0082, 0.0187]$; wins 10/10; Wilcoxon/sign $p < 0.001$.
Shared task–seed pairs	$\Delta = 0.0134$, CI $[-0.0021, 0.0302]$; wins 10/40; exposes floor-task heterogeneity.
Task-wise non-floor	$\Delta = 0.0580$, CI $[0.0358, 0.0814]$; wins 18/20; Wilcoxon $p < 0.001$.
Sunspots gate	$\Delta = 0.0242$, CI $[0.0026, 0.0565]$; wins 8/10; Wilcoxon $p = 0.0420$; sign $p = 0.0547$.

IV. DISCUSSION AND SCOPE

The accuracy comparison is fair for a narrow benchmark statement: tasks, splits, seeds, ridge grid, holdout protocol, paired keys, and classical readout dimension are aligned; both models are selected without holdout access. It is not a full resource comparison. QRC96 requires many Pauli observables, and finite-shot sampling, measurement-basis scheduling, calibration, and hardware noise are outside this simulated study. The reviewer-safe interpretation is therefore: the phase maps identify a reproducible memory-defined operating regime; within that regime, QRC96 features are competitive with, and on NARMA10/Sunspots outperform, an audited ESN100 at matched readout dimension.

V. CONCLUSION

Dissipative QRC performance is organized by a memory-defined operating regime. Phase maps transfer across tasks and seeds, memory diagnostics predict the useful region, and ablations show that recurrent mixing plus controlled damping are necessary before richer readout access helps. The ESN

Supporting evidence: paired gains, task-wise deltas, and mechanisms

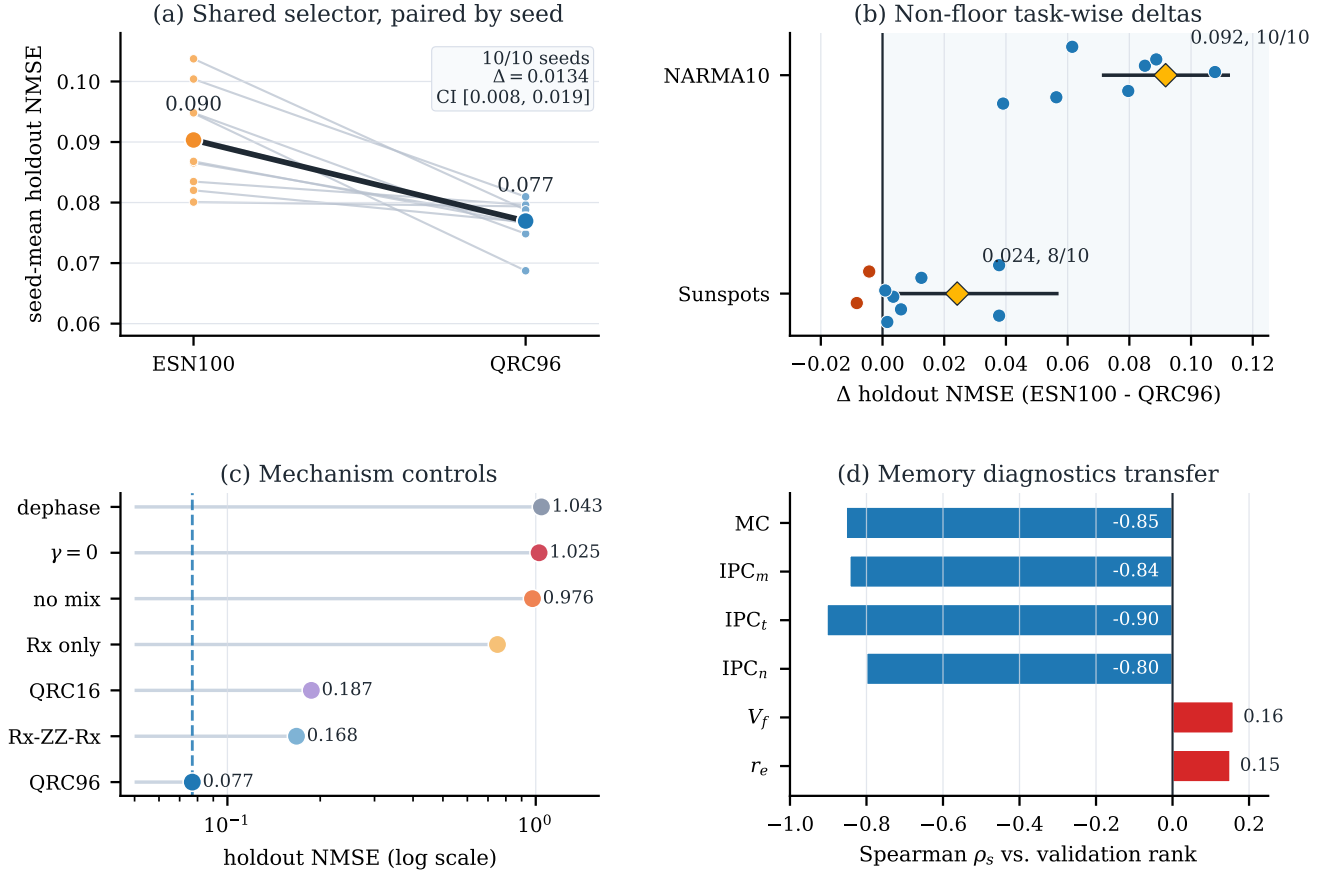


Fig. 2. Supporting evidence for the operating-regime claim. (a) At comparable readout dimension, shared QRC96 improves seed-level mean holdout NMSE over ESN100. (b) Under validation-plateau task-wise selection, the same QRC96 architecture improves NARMA10 and Sunspots. (c) Mechanism controls show that richer readout helps only when recurrent mixing and nonzero damping preserve usable memory. (d) Memory and IPC diagnostics predict validation rank, whereas diversity-only diagnostics do not.

TABLE II
TASK-WISE VALIDATION-PLATEAU HOLDOUT COMPARISON. POSITIVE Δ FAVORS QRC96. SETTINGS ARE SELECTED BY VALIDATION ONLY AND WRITTEN AS QRC ($\beta/\pi, \lambda/\pi, \gamma$) AND ESN (ρ , input, leak).

Task	ESN100	QRC96	Δ	QRC wins	QRC setting	ESN setting
Mackey–Glass	7.7e-07	9.6e-06	-8.8e-06	0/10	0.030/0.125/0.080	0.9/0.1/1.0
Lorenz	3.3e-09	2.4e-06	-2.4e-06	0/10	0.030/0.125/0.120	0.5/0.5/1.0
NARMA10	0.2161	0.1244	0.0918	10/10	0.020/0.125/0.080	0.9/0.1/1.0
Sunspots	0.2062	0.1820	0.0242	8/10	0.010/0.040/0.095	0.5/0.1/0.3

comparison supports, but does not replace, the phase-map claim: at matched classical readout dimension and holdout-blind validation selection, the same QRC96 architecture gives strong aggregate accuracy and robust non-floor NARMA10/Sunspots wins. The result is not a quantum-advantage or hardware-resource claim; the next tests are finite measurements, calibration noise, and expanded classical baselines.

REFERENCES

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TABLE III

ARTIFACT MAP FOR EXACT REPRODUCTION. THE PAPER PDF, FIGURES, AND TABLES ARE GENERATED FROM CHECKED-IN CSV/JSON ARTIFACTS; HOLDOUT DATA ARE READ ONLY AFTER VALIDATION SELECTION.

Artifact	Role
QRC16 grid	data/qrc_seed_ensemble_grid.csv: coarse grid for regime discovery and leave-one-task-out phase-map transfer.
QRC96 grid	data/qrc96_same_arch_expanded_grid.csv: same-architecture validation search with 125 configurations, ten seeds, and four tasks.
Sunspots audit	data/qrc96_sunspots_fine_refinement_grid.csv: same-architecture refinement used only by the task-wise validation-plateau selector.
ESN grid	data/esn_seed_ensemble_grid.csv: audited ESN16/ESN100 sweep with the same tasks, splits, seeds, and ridge grid.
Statistics	data/qrc96_esn100_stats.json and data/qrc96_esn100_taskwise_stats.json: selected configs, paired deltas, bootstrap CIs, Wilcoxon tests, sign tests, and gate checks.
Selectors	scripts/analyze_qrc96_esn100*.py: validation-only selection with matched task/seed keys enforced before holdout comparisons.
Figures	scripts/make_figures_and_build_data.py: regenerates all plotted values and figure files used by the manuscript.
Build	./reproduce.sh and ./paper/build.sh --update-pdf: end-to-end regeneration of data products and the submitted PDF.

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